

REASONING MASTERCLASS: CLOCK

Comprehensive Theory, 15 Solved Examples & 25 Practice Questions with Answer Key

1. Introduction & Core Concepts (Theory)

In logical reasoning, problems based on **Clocks** are extremely common. A standard clock is a circular dial that spans 360° and is divided into 12 equal hour divisions. Each hour space represents an angle of 30° ($360^\circ / 12 = 30^\circ$).

Important Facts & Speeds:

- **Minute Hand Speed:** The minute hand covers 360° in 60 minutes, which equals $360^\circ / 60 = 6^\circ \text{ per minute}$.
- **Hour Hand Speed:** The hour hand covers 360° in 12 hours (or 30° in 60 minutes), which equals $30^\circ / 60 = 0.5^\circ \text{ per minute}$ (or $1/2^\circ \text{ per minute}$).
- **Relative Speed:** The minute hand gains $6^\circ - 0.5^\circ = 5.5^\circ$ or $11/2^\circ \text{ per minute}$ over the hour hand.

Angle Between Hands Formula

The angle (θ) between the hour and minute hands at any given time (H hours and M minutes) is calculated using the formula:

$$\theta = \left| 30H - \frac{11}{2}M \right|$$

Note: If the calculated angle exceeds 180° , the reflex angle is obtained by subtracting it from 360° .

Key Conditions & Shortcuts

- **Coincidence (0° angle / Overlapping):** Both hands are pointing in the same direction. This occurs 11 times in 12 hours and 22 times in 24 hours. Formula: $\text{ext}\{Time\} = \frac{2}{11} \text{imes } (\text{ext}\{H\} \text{imes } 30)$ (for the given hour H).
- **Opposite (180° angle / Straight line):** Both hands are pointing in exact opposite directions. Occurs 22 times in a day.
- **Right Angle (90° angle):** Both hands are perpendicular to each other. Occurs twice per hour, 44 times in 24 hours.
- **Faulty Clocks (Gain/Loss):** If a clock runs faster or slower, the ratio method is applied: $\text{ext}\{Correct\ Time\} = \text{ext}\{Wrong\ Time\} \text{imes } \frac{\text{ext}\{Total\ Correct\ Hours\}}{\text{ext}\{Total\ Faulty\ Hours\}}$.

2. Solved Examples (15 Examples)

Example 1: Find the angle between the hands of a clock at 4:40 PM.

Solution: Given $H = 4$, $M = 40$.

Formula: $\text{heta} = |\text{left}| 30(4) - \text{rac}{11}{2}(40) \text{ ight}| = |120 - 220| = |-100| = 100^\circ$.

Answer: 100°

Example 2: At what time between 7 and 8 o'clock do the hands coincide?

Solution: For coincidence between H and (H+1) o'clock, formula is $\text{rac}{2}{11} \text{ imes } (H \text{ imes } 30)$.

Here $H = 7$: $\text{rac}{2}{11} \text{ imes } (7 \text{ imes } 30) = \text{rac}{420}{11} = 38 \text{ rac}{2}{11}$ minutes past 7.

Answer: 7 hours $38 \text{ rac}{2}{11}$ minutes.

Example 3: At what time between 3 and 4 o'clock will the hands be in opposite directions (180°)?

Solution: For opposite directions, add 6 hours to the base hour if $H < 6$ ($3 + 6 = 9$).

$\text{rac}{2}{11} \text{ imes } (9 \text{ imes } 30) = \text{rac}{540}{11} = 49 \text{ rac}{1}{11}$ minutes past 3.

Answer: 3 hours $49 \text{ rac}{1}{11}$ minutes.

Example 4: At what times between 5 and 6 o'clock are the hands at right angles (90°)?

Solution: Right angles happen 15 minute-spaces apart. For 5 o'clock base:

Case 1 ($5 - 15 \text{ ext{ min}} = 2 \text{ ext{ o'clock base}}$): $\text{rac}{2}{11} \text{ imes } 60 = 10 \text{ rac}{10}{11}$ min past 5.

Case 2 ($5 + 15 \text{ ext{ min}} = 8 \text{ ext{ o'clock base}}$): $\text{rac}{2}{11} \text{ imes } 240 = 43 \text{ rac}{7}{11}$ min past 5.

Answer: 5h $10 \text{ rac}{10}{11}$ m and 5h $43 \text{ rac}{7}{11}$ m.

Example 5: A clock loses 10 minutes every 24 hours. If it is set right on Sunday 10 AM, what time will it show on Tuesday 10 AM?

Solution: Total hours from Sunday 10 AM to Tuesday 10 AM = 48 hours.

Total time lost in 48 hours = $2 \text{ imes } 10 = 20$ minutes.

Correct time - 20 minutes = Tuesday 10:00 AM - 20 mins = 9:40 AM.

Answer: Tuesday 9:40 AM.

Example 6: Find the angle between the hands of a clock at 10:15.

Solution: $H = 10$, $M = 15$.

$$\theta = |\text{left}| 30(10) - \text{right}| \frac{11}{2}(15) \text{ right}| = |300 - 82.5| = 217.5^\circ.$$

$$\text{Reflex angle: } 360^\circ - 217.5^\circ = 142.5^\circ.$$

Answer: 142.5° (or $142 \frac{1}{2}^\circ$)

Example 7: The mirror image of a clock shows 8:20. What is the actual time?

Solution: Subtract the given time from 11:60 (or 23:60).

$$11:60 - 8:20 = 3:40.$$

Answer: 3:40

Example 8: The water image of a clock shows 4:42. Find the actual time.

Solution: For water image, subtract from 17:90 (since minutes > 30):

$$17:90 - 4:42 = 13:48, \text{ which is } 1:48 \text{ PM.}$$

Answer: 1:48

Example 9: How many times in 24 hours do the hands of a clock overlap?

Solution: They overlap 11 times in 12 hours. Therefore, in 24 hours, $11 \times 2 = 22$ times.

Answer: 22 times

Example 10: At what time between 2 and 3 o'clock do the hands coincide?

Solution: $\frac{2}{11} \times (2 \times 30) = \frac{120}{11} = 10 \frac{10}{11}$ minutes past 2.

Answer: 2 hours $10 \frac{10}{11}$ minutes.

Example 11: At what time between 8 and 9 o'clock are the hands opposite to each other?

Solution: $8 - 6 = 2$ o'clock base. $\frac{2}{11} \times (2 \times 30) = 10 \frac{10}{11}$ minutes past 8.

Answer: 8 hours $10 \frac{10}{11}$ minutes.

Example 12: A clock gains 3 minutes every hour. If set right at 12 noon, what time will it show at 9 PM?

Solution: Total elapsed time = 9 hours. Total gain = $9 \times 3 = 27$ minutes.

Shown time = 9:00 PM + 27 mins = 9:27 PM.

Answer: 9:27 PM

Example 13: How many times do the hands form a right angle in 12 hours?

Solution: 22 times (24 times in a full day, with 44 total).

Answer: 22 times

Example 14: Through how many degrees does the minute hand turn in 12 minutes?

Solution: $12 \times 6^\circ = 72^\circ$.

Answer: 72°

Example 15: Find the angle between the hands at 12:35.

Solution: $H = 0, M = 35$.

$\theta = \left| 0 - \frac{11}{2}(35) \right| = 192.5^\circ$.

Minor angle: $360^\circ - 192.5^\circ = 167.5^\circ$.

Answer: 167.5°

3. Practice Questions (25 Questions)

Q1. What is the angle between the hands of a clock at 3:20?

(A) 20°

(B) 25°

(C) 30°

(D) 35°

Q2. At what time between 9 and 10 o'clock are the hands opposite to each other?

(A) 9h 16 $\frac{4}{11}$ m

(B) 9h 27 $\frac{3}{11}$ m

(C) 9h 32 $\frac{8}{11}$ m

(D) 9h 45m

Q3. A clock's mirror image reads 4:40. What is the actual time?

- (A) 7:20 (B) 8:20
(C) 7:40 (D) 6:20

Q4. A clock shows 4:10. What time does its water image show?

- (A) 2:20 (B) 1:50
(C) 2:50 (D) 3:20

Q5. At what time between 1 and 2 o'clock do the hands coincide?

- (A) 1h 5 $\frac{5}{11}$ m (B) 1h 6 $\frac{6}{11}$ m
(C) 1h 10m (D) 1h 7 $\frac{5}{11}$ m

Q6. Through how many degrees does the hour hand move in 1 hour?

- (A) 30° (B) 6°
(C) 360° (D) 15°

Q7. At what time between 6 and 7 o'clock are the hands at right angles?

- (A) 6h 16 $\frac{4}{11}$ m (B) 6h 32 $\frac{8}{11}$ m
(C) Both (A) and (B) (D) None of these

Q8. A clock gains 15 minutes in 24 hours. If set right at 2 PM, what is the true time when the clock indicates 4 PM 4 days later?

- (A) 3:00 PM (B) 3:40 PM
(C) 3:30 PM (D) 4:00 PM

Q9. What is the angle between the hands at 2:50?

- (A) 110° (B) 115°
(C) 125° (D) 135°

Q10. How many times in 24 hours are the hands in opposite directions?

- (A) 20 (B) 22
(C) 24 (D) 44

Q11. At what time between 10 and 11 o'clock do the hands coincide?

- (A) 10h 50m (B) 10h 54 $\frac{6}{11}$ m
(C) 10h 45 $\frac{5}{11}$ m (D) 10h 58m

Q12. The time in a clock is 7:35. What is its mirror image?

- (A) 4:25 (B) 5:25
(C) 4:35 (D) 3:25

Q13. What is the angle between the hands at 5:24?

- (A) 12° (B) 15°
(C) 18° (D) 24°

Q14. A clock shows 8:20. Find its water image.

- (A) 10:10 (B) 9:40
(C) 10:40 (D) 11:10

Q15. At what time between 11 and 12 o'clock are the hands opposite?

- (A) 11h 27 $\frac{3}{11}$ m (B) 11h 32 $\frac{8}{11}$ m
(C) 11h 6 $\frac{6}{11}$ m (D) Exactly 12:00

Q16. Through how many degrees does the minute hand move in 40 minutes?

- (A) 200° (B) 220°
(C) 240° (D) 250°

Q17. At what time between 4 and 5 o'clock do the hands coincide?

- (A) 4h 21 $\frac{9}{11}$ m (B) 4h 20m
(C) 4h 22 $\frac{2}{11}$ m (D) 4h 25m

Q18. How many right angles do the clock hands form in a week (7 days)?

- (A) 308 (B) 336
(C) 316 (D) 322

Q19. What is the angle between the hands at 12:10?

- (A) 50° (B) 55°
(C) 60° (D) 65°

Q20. If the actual time is 11:20, what is the mirror image time?

- (A) 1:20 (B) 12:40
(C) 1:40 (D) 12:20

Q21. At what time between 7 and 8 o'clock are the hands at right angles?

- (A) 7h 21 $\frac{11}{11}$ m (B) 7h 54 $\frac{6}{11}$ m
(C) Both (A) and (B) (D) None of these

Q22. A clock gains 10 seconds every 3 hours. If set right at 12 noon, how much ahead will it be at 12 noon next day?

- (A) 40 seconds (B) 60 seconds
(C) 80 seconds (D) 90 seconds

Q23. What is the angle between the hands at 1:45?

- (A) 135.5° (B) 142.5°
(C) 140° (D) 150°

Q24. Through how many degrees does the hour hand move in 36 minutes?

- (A) 15° (B) 18°
(C) 20° (D) 24°

Q25. At what time between 3 and 4 o'clock are the hands 3 minute-spaces apart?

- (A) 3h 6m (B) 3h 27 $\frac{3}{11}$ m
(C) Both (A) and (B) (D) None of these

4. Practice Questions Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
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1	(B)	6	(A)	11	(B)	16	(C)	21	(C)
2	(B)	7	(C)	12	(A)	17	(A)	22	(C)
3	(A)	8	(B)	13	(C)	18	(A)	23	(B)
4	(B)	9	(B)	14	(C)	19	(B)	24	(B)
5	(A)	10	(B)	15	(A)	20	(C)	25	(C)